

$\Sigma(0, 4, 3)$

Exercises 1.9

Obtain disjunctive normal form, conjunctive normal, principal disjunctive normal form and principal conjunctive normal form for the following

$$\begin{aligned} 1. Q \wedge (P \vee \neg Q) &\equiv (Q \wedge P) \vee (Q \wedge \neg Q) \text{ (2nd Distributive Law)} \\ &\equiv (Q \wedge P) \vee F \text{ (2nd Negation Law)} \\ &\equiv (Q \wedge P) \text{ (2nd Identity Law)} \end{aligned}$$

Conjunctive Normal Form: $(Q \wedge P)$

$$\begin{aligned} &\rightarrow \equiv ((Q \vee F) \wedge (P \vee F)) \text{ (2nd Identity Law)} \\ &\equiv ((Q \vee (P \wedge P)) \wedge (P \vee (Q \wedge \neg Q))) \text{ (2nd Negation Law)} \\ &\equiv ((Q \vee P) \wedge (Q \vee \neg P)) \wedge ((P \vee Q) \wedge (P \vee \neg Q)) \text{ (2nd Distributive Law)} \end{aligned}$$

Principal

Conjunctive Normal form: $[(P \vee Q) \wedge (\neg P \vee Q) \wedge (P \vee \neg Q)]$

$\Pi(0, 1, 2)$

$$Q \wedge (P \vee \neg Q) \equiv (Q \wedge P) \equiv (P \wedge Q) \text{ (2nd Commutative Law)}$$

Principal Disjunctive Normal form: $(P \wedge Q)$

$\Sigma(3)$